

DevOps		
Objective: To gain a comprehensive understanding of DevOps principles and practices, including automation, collaboration, and continuous integration/deployment. Develop proficiency in Linux, Bash scripting, Git, containerization with Docker, and cloud computing with AWS. Enhance skills in managing scalable and reliable applications using modern DevOps tools and methodologies. Prepare for hands-on implementation of DevOps workflows for streamlined software development and deployment.		
Duration:		
S.NO	CONTENT	HOURS (L/P)
	Introduction to DevOps	
1	Introduction to DevOps:	
	Streamlining software development through collaboration	
	Automation	
	Continuous integration/deployment	
2	What is SDLC:	
	Software Development Life Cycle	
	Planning	
	Creating	
	Testing	
	Deploying software systems	
3	Benefits of DevOps:	
	Efficiency	
	Automation	
	Faster deployment	
	Scalability	
	Reliability	
	Continuous integration	
	Delivery	
4	Virtual Machines:	
	Emulated computers running isolated environments with resource configuration through a hypervisor.	
	Linux	
5	Linux Basics:	
	Open-source OS – commands	
	File management	
	Permissions	
	Networking	
6	Linux Process:	
	Managing resources and multitasking in a secure and scalable	
7	Users and Groups:	
	Define file access	
	Determine read/write/execute permissions	
8	Linux Commands	
	Basics like mkdir, cp, mv, touch, etc.	

	Bash Scripting	
9	Introduction to Bash Scripting: Automating tasks in Linux/Unix environments	
10	Scripts and Variables: Execute predefined actions Use variables to store data, modify behavior with command-line arguments	
11	User Inputs: Guide program behavior based on input for interactive systems	
12	Loops: Automate repetitive tasks using loops like while	
	Git	
13	Git Basics: Version control to track code changes Manage repositories Commit Collaborate Manage code history	
14	Benefits of Git: Efficient collaboration Version control Branching Merging	
15	Git Commands: Staging/unstaging Branching Merging Rebasing	
16	Git SSH Login: Secure authentication to communicate with remote repositories.	
	Introduction to Containers	
17	Containers: Lightweight Portable Isolated environments for running applications	
18	Docker: Platform to develop Ship Run containerized applications	
19	Hands-on Docker: Deploy Scale apps using containers	
	Microservices:	

20	Independent	
	Scalable components for modular development	
21	Docker Images:	
	Containers with app dependencies for efficient deployment	
22	Docker Volumes & Networking:	
	Managing networks and volumes across containers	
AWS Essentials		
23	Cloud Computing:	
	Scalable	
	Flexible	
	Infrastructure	
	Services on-demand	
24	AWS Overview:	
	Cloud platform offering computing power	
	Storage	
	Services	
25	AWS in DevOps:	
	Automate infrastructure	
	CI/CD	
	Monitoring with AWS	
26	Advantages of AWS:	
	Scalability	
	Reliability	
	Flexibility	
	Security	
	Global reach	